gpeyt1-30 · Layered k (vertical)

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A 12 ft soil column beneath a detention pond consists of 4 ft of sand (k = 2×10^{-3} cm/s), 2 ft of silty clay (k = 5×10^{-6} cm/s), and 6 ft of silty sand (k = 8×10^{-4} cm/s), top to bottom. The total head loss across the column is 9 ft. Flow is vertical (downward).

Most nearly, the equivalent vertical hydraulic conductivity is:

A. 1.1e-03 cm/s
B. 2.9e-05 cm/s
C. 2.0e-04 cm/s
D. 5.0e-06 cm/s

Given equations / data

> kv,eq = ΣHi/Σ(Hi/ki); kh,eq = Σ(kiHi)/ΣHi

Worked solution

1. Σ Hi/ki = 4/2e-3 + 2/5e-6 + 6/8e-4 = 409,500 ft·s/cm
2. kv = 12/409,500 = 2.93e-05 cm/s
3. (kh = 1.07e-03 cm/s for comparison; clay layer controls vertical flow)

Final answer

2.9e-05 cm/s

Why the other values are wrong

- **1.1e-03 cm/s** — used the horizontal (arithmetic) average
- **2.0e-04 cm/s** — used a geometric mean
- **5.0e-06 cm/s** — took the clay k alone

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.3 Flow Through Soil (layered/equivalent k), p. 188

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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